

Antoine COMMARET

Post-doctorate - 2026

Nice, France

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📄 <https://acommaret.github.io>



📖 Education

- 2025 – 2026 **Post-doctorate**, INRIA Sophia-Antipolis, ABS Team, Valbonne, France.
Stratification of point clouds.
Optimization of non-smooth functions.
- 2021 – 2024 **PhD Candidate**, DataShape Team, INRIA Sophia-Antipolis & J. A. Dieudonné Lab, France.
"Persistent Geometry", under the supervision of David Cohen-Steiner & Indira Chatterji.
- 2019 – 2020 **Master 2 "Mathématiques, Vision, Apprentissage" with highest honors**, ENS Paris-Saclay.
Computational geometry and statistics courses.
- 2017 – 2019 **Master 1 & Last Year of Bachelor**, École Normale Supérieure de Paris, France.
General advanced maths in the highly selective mathematics path of the ENS.
- 2015 – 2017 **Classes préparatoires aux Grandes Ecoles**, Lycée Hoche, Versailles, France.
Admitted to both École Polytechnique and ENS Paris after 2 years of intense preparation.

📄 List of publications

Authors are listed alphabetically and contributed equally to each paper.

- 2024 David Cohen-Steiner, Antoine Commaret. *Persistent Intrinsic Volumes*. Published in *Advances in mathematics*, Volume 503, Article 111195 (2026). [Link](#).
- 2024 Antoine Commaret. *Generalized Morse theory for tubular neighborhoods*. Accepted pending revisions in the *Journal of Topology and Analysis*. [arXiv link](#).
- 2025 Frédéric Cazals, Antoine Commaret, Louis Goldenberg. *Modeling high dimensional point clouds with the spherical cluster model*. Preprint, [arXiv link](#).
- 2026 David Cohen-Steiner, Antoine Commaret. *Continuity of the normal cycle with respect to C^0 -Topologies*. Preprint, [arXiv link](#).

👛 Research residency and internships

- July 2025 **Research guest**.
Institute of Geometry, directed by Michael Kerber. *Graz, Austria*.
Charles University. Visiting Jan Rataj. *Prague, Czechia*.
Algebraic Systems Biology team directed by Heather Harrington. *Dresden, Germany*.
- Feb 2023 **Startup Week 3IA**, Sophia-Antipolis.
One week project on the segmentation femoral arteries on ultrasound scans for Caranx Medical.
- June 2022 **Summer School EUTOPIA**, Paris.
Topology in the context of soft matter and proteins.
- 2020 – 2021 **Research Internship**, INRIA Saclay, under the supervision of Frédéric Chazal & Bertrand Michel.
TDA and study of the dimension reduction algorithm UMAP.
- 2019 **Research Internship**, Göttingen University, Germany, under the supervision of Harald Helfgott.
Analytic Number Theory. Reviewing the proof of the weak Goldbach conjecture.

🏛 Teaching

- 2026 **Tutor in geometry**, ABS team.
Tutor in differential geometry for research engineers in my INRIA team.
- 2021 – 2026 **Lecturer**, Université Côte d'Azur.
Introduction to Topological Data Analysis (M2), Graph Theory, Analysis I, Algebra II (Bachelor)
321 hours in total.
- 2017 – 2021 **Colleur in Preparatory Classes**, Lycée Hoche, Versailles.
Weekly oral exams.



Future talks

- Jan 2027 **Geometry seminar**, *LAREMA Lab, Angers, France.*
Nov 2026 **AATRN seminar**, *online.*
Oct 2026 **Integral Geometry Day**, *Sorbonne Université, Paris, France.*



Talks & conferences

- July 2026 **Foundation of Computational Mathematics Workshop**, *Vienna, Austria.*
Talk on the application of TDA in Geometric Inference
June 2026 **Curves and Surfaces Conference**, *Saint-Malo, France.*
Talk on the application of TDA in Geometric Inference
Feb 2026 **Geometry and topology seminar of Sorbonne Université**, *Paris, France.*
Talk on the connections between integral geometry and persistent homology.
July 2025 **Charles University and Max-Planck Institute**, *Prague, Czechia, and Dresden, Germany.*
Introduction to persistent geometry.
March 2025 **Jean Kutzmann lab seminar**, *Grenoble, France.*
Talk about PhD results.
July 2024, **Young Topologist Meeting**, *Münster, Germany, and Lausanne, Switzerland.*
July 2023 Talks for *Generalized Morse theory for tubular neighborhoods* and *Persistent Intrinsic Volumes.*
Oct 2023 **"Young Researcher in Geometry" Day**, *Paris, France.*
Introduction to Geometric measure theory.
Feb 2023 **World AI Festival**, *Cannes, France.*
Presenting the work done during the startup week 3IA.
June 2022 **Math PhD Students' Colloquium**, *Nice, France.*
& 2023 Talks about persistent homology & Organization.



Organisation

- 2027 **Climathics**, *CIRM, Marseille, France.*
2023, 2025 **Journées Jeunes Chercheurs en géométrie**, *Paris, France.*
2022-2024 **Organizer of the regular seminar of the Datashape team**, *Sophia-Antipolis, France.*
June 2023 **Workshop for PhD students of Université Côte d'Azur**, *Fréjus, France.*



Languages

- Native French
Fluent English, German
Beginner Esperanto, Italian



Softwares

- Writing L^AT_EX
Code Numpy, Scipy, Pytorch, GUDHI libraries.
Graphics Inkscape



Miscellaneous

- Tenor in the Université Côte d'Azur Choir
Singer and bassist of the progressive rock band *Fondamental*
Interests in History, Geography, Litterature and Biology